

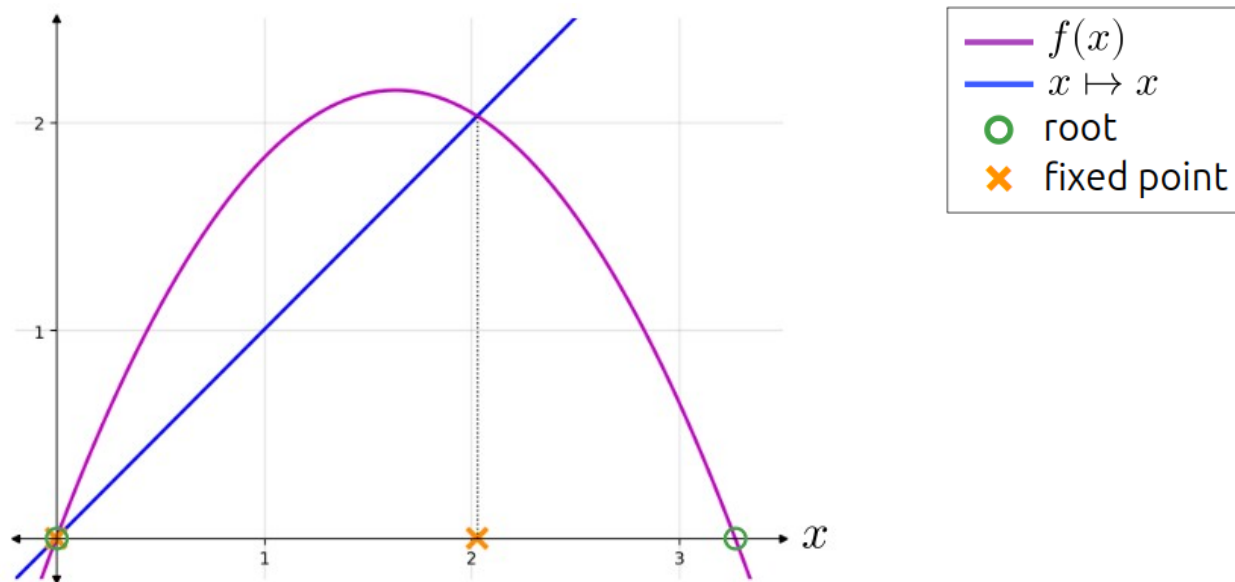
E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 12-B

Fixed points of functions

Terminology

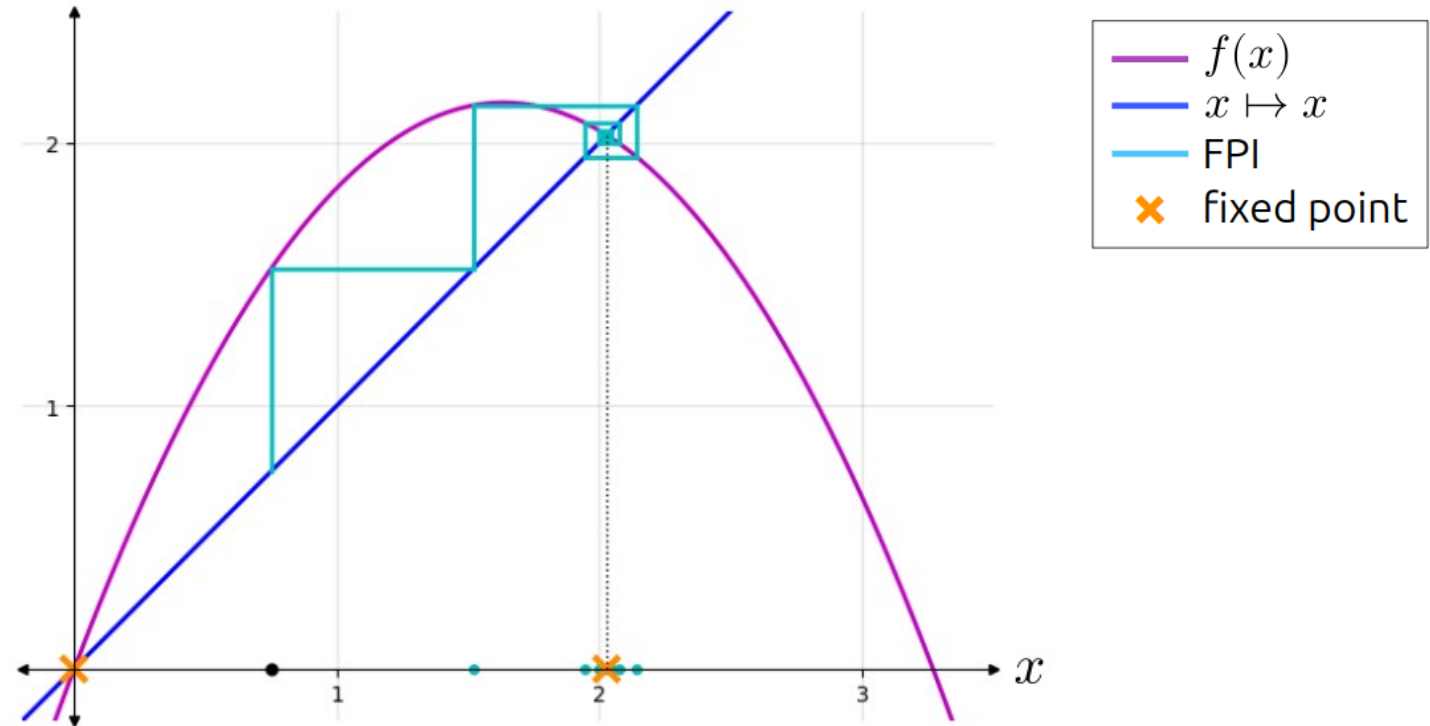
- A **fixed point** of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a point that maps to itself.
- That is, x^* is a fixed point of f if $f(x^*) = x^*$



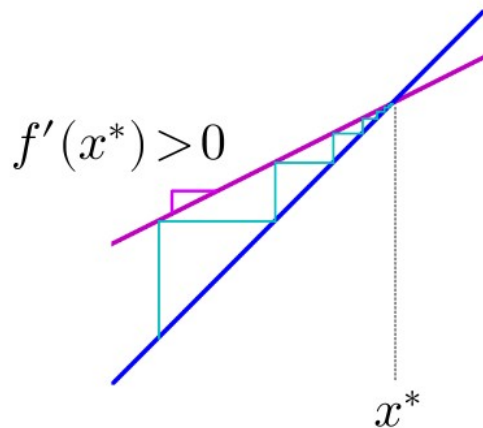
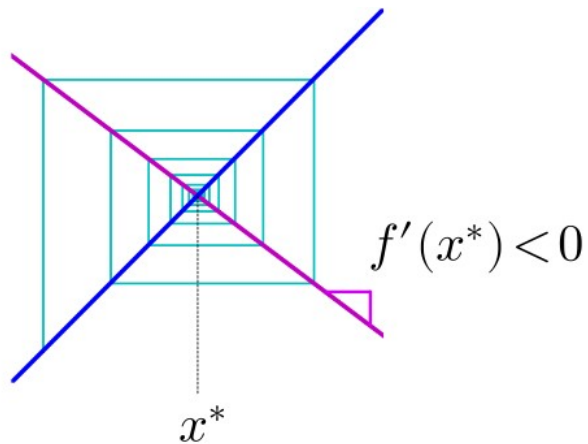
- A ***fixed point iteration*** (FPI) of is a sequence generated with,

choose x_0

$$x_{k+1} = f(x_k)$$



Convergence of FPI: Scalar functions



- linear case: $f(x) = ax + b$

FPI converges to $x^* = b/(1 - a)$ for any x_0 if and only if $|a| < 1$.

- nonlinear case: $f(x) : \mathbb{R} \rightarrow \mathbb{R}$

FPI converges to $x^* \in [a, b]$ for any $x_0 \in [a, b]$ if and only if $\max_{x \in [a, b]} |f'(x)| < 1$.

Using FPI to solving nonlinear equations

- We wish to find a root of $f(x)$.
- Apply FPI to a function $g(x)$ whose fixed points are roots of $f(x)$

$$g(x^*) = x^* \quad \Rightarrow \quad f(x^*) = 0$$

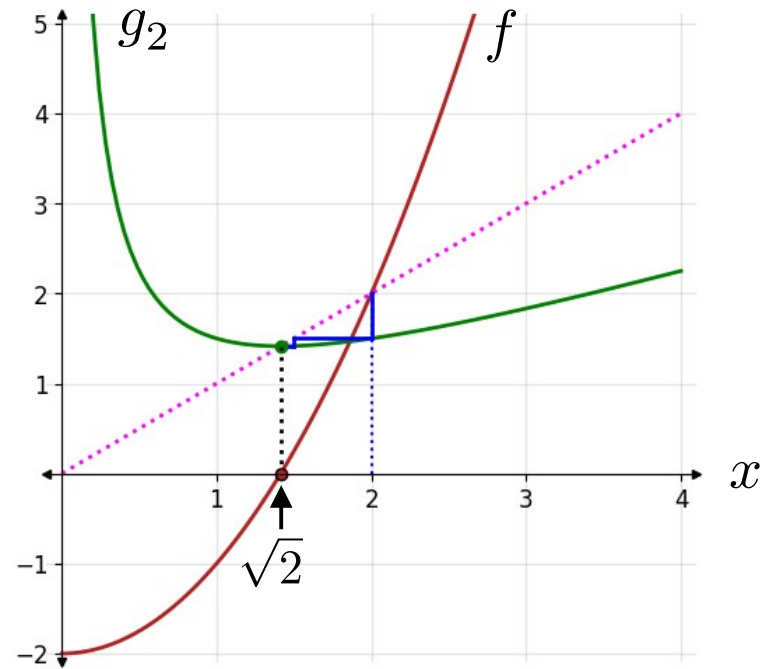
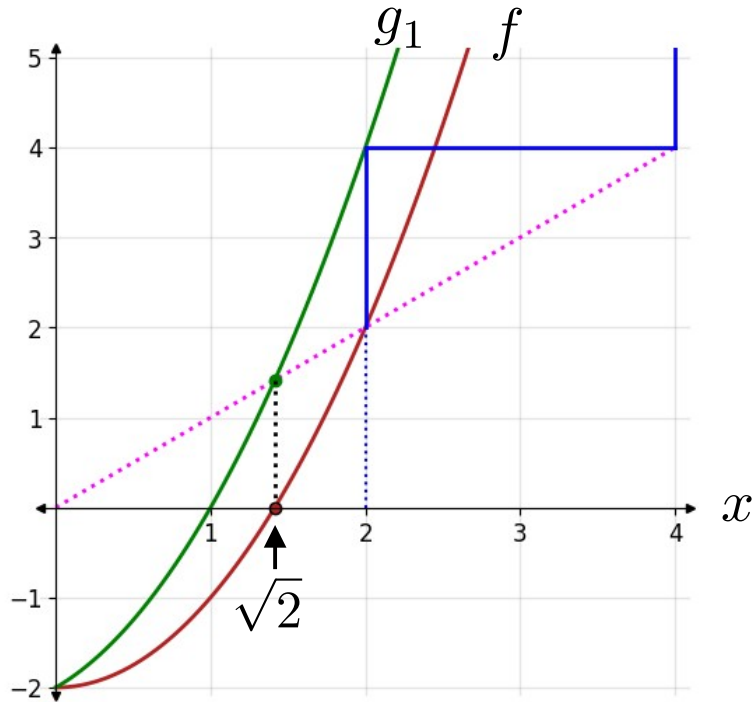
- How to choose $g(x)$?
- $g(x) = f(x) + x$ always works, but there may be better options.
- The shallower, the better.

Example: Find the positive root of $f(x) = x^2 - a$

$$g_1(x) = f(x) + x$$

$$g_2(x) = \frac{1}{2} \left(\frac{a}{x} + x \right)$$

e.g. $a = 2$



Using FPI to solve systems of nonlinear equations

- f is a system of n nonlinear equations in n unknowns. $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$
- Suppose R is some given region in $\mathbb{R}^n$
- Theorem:

If f has a fixed point in R , then FPI will converge for any $x_0 \in R$ provided all partial derivatives of f have absolute value less than 1 everywhere in R .

- This is a *conservative* statement, meaning that FPI *may* converge even when some of the partial derivatives are larger than 1.

Example: Solve
$$\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$$

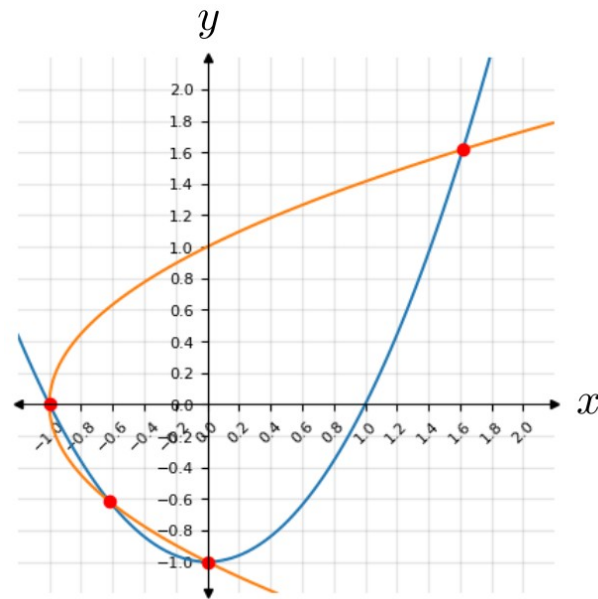
- Find the roots of

$$f : (x, y) \mapsto (x^2 - y - 1, y^2 - x - 1)$$

- Two options for FPI:

$$g : \begin{cases} x = y^2 - 1 \\ y = x^2 - 1 \end{cases} \Rightarrow \begin{aligned} g_1(x, y) &= y^2 - 1 \\ g_2(x, y) &= x^2 - 1 \end{aligned}$$

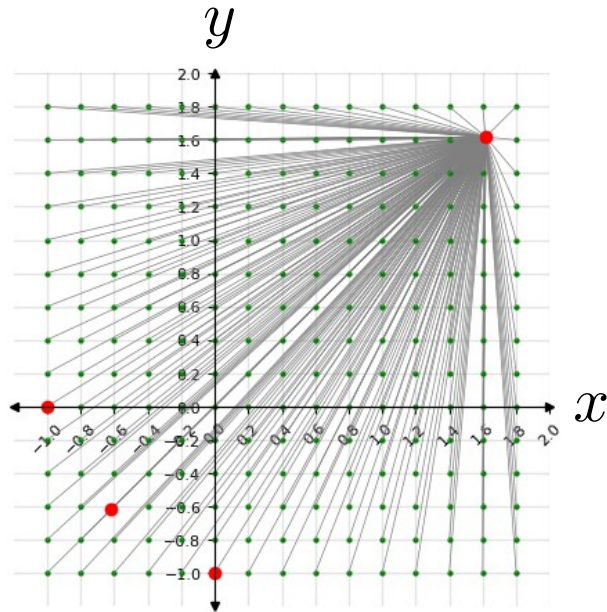
$$h : \begin{cases} x = \sqrt{1 + y} \\ y = \sqrt{1 + x} \end{cases} \Rightarrow \begin{aligned} h_1(x, y) &= \sqrt{1 + y} \\ h_2(x, y) &= \sqrt{1 + x} \end{aligned}$$



- Convergence analysis

- Perform FPI over a grid of initial conditions.
- Green dots are initial conditions for which FPI converged.
- Gray lines point to the fixed points to which it converged.

FPI on g_1



FPI on g_2

